SERIE N · Autumn Railways

UI blueprint & implementation guide — v1.0.0

For the implementing developer (Sol). Everything needed to replace the current prototype interface with the new brass-and-enamel UI, without guessing.



Package: serien-ui-package/ · Reference: reference/serien-ui-reference.html · Spec: spec/ui-spec.json · Tokens: tokens/ · CSS: css/serien-ui.css · Library: library/serien-ui-library.html
Date: 6 November 2026

1 · Read this first

Golden rules (do not skip)

1. **The reference HTML is the source of truth.** If this PDF, the JSON spec and the reference disagree, follow `reference/serien-ui-reference.html` and report the mismatch.
2. **Do not redesign or "clean up" the CSS.** `css/serien-ui.css` is order-dependent: later rules refine earlier ones. Load `tokens/tokens.css` then `css/serien-ui.css`, unchanged.
3. **Keep every class name and id** used in the reference DOM. CSS and behaviour depend on them (list in section 6 and `spec/ui-spec.json → components`).
4. **The 3D layout view is NOT part of this package.** The engine canvas goes inside `.win`; the UI only adds overlays on top.
5. **Never hard-code 3 trains or 7 switches.** Render from game data: 0–8 trains, N switches.
6. **The reference contains a mock simulation.** Replace the parts marked MOCK (section 8) with real game state; port the parts marked KEEP as they are.
7. **Switch IDs:** the reference uses IDs derived from the track drawing (WX-A, WX-B, Y-IN, Y-1, Y-2, Y-R1, Y-R2). The current game uses WX-A, WX-B, Y-IN, Y-2, Y-3, Y-R1, Y-R2, Y-4. **Use the game's IDs and descriptions**; the panel is generated from the game registry.

Package contents

Path	What it is
`reference/serien-ui-reference.html`	Self-contained working reference (mock simulation). Open in Chrome. Query param ?trains=0..8 previews any train count.
`css/serien-ui.css`	Production stylesheet, 1:1 with reference. ORDER-DEPENDENT. Do not reorder/split.
`tokens/tokens.css`	Fonts @import + design tokens (--sn-*) + core shorthands. Load first.
`tokens/tokens.json`	Same tokens + behaviour constants (thresholds, timings).
`assets/svg/serien-sprite.svg`	Gradients, icons, loco silhouettes, edition emblem. Inline once in <body>.
`assets/textures/*.jpg`	Brushed brass, enamel, brushed steel, mahogany, parchment, spun brass (gauge bezels).
`library/serien-ui-library.html`	Component gallery with every state, live CSS.
`spec/ui-spec.json`	This file.
`docs/SerieN-UI-Blueprint.pdf`	Human blueprint + migration guide from current prototype.
`screenshots/`	Reference screenshots used in this PDF (00 = current prototype).

How to look at the reference

- Open `reference/serien-ui-reference.html` in Chrome (desktop) with internet access (Google Fonts). Add `?trains=0` … `?trains=8` to preview any number of trains.
- Try: Environment slider (day → sunset → night), throw WX-B to Diverging and run BR 216 to trigger a collision alarm, press STOP ALL twice, hold Reset, open Menu / Track plan / Help, resize to tablet and phone.
- Menu → Graphics quality → High to see full materials (Auto may pick Low on slow machines).

Fonts in the screenshots

Screenshots in this PDF were rendered on a machine without access to Google Fonts, so text appears in a fallback serif. The real UI uses **Cinzel Decorative 700** (display), **Federo** (labels) and **Cutive Mono** (data) — all SIL OFL, loaded by `tokens.css`. Self-host them for production. Sizes, colours and layout in the screenshots are correct.

2 · Current prototype → new UI (visual mapping)

Numbers match the table on the next page. A number missing from the new UI means the element was **removed** or **moved into a dialog**.

Current prototype (to be replaced)



New UI (1440 × 900)



S = new Sound toggle. 8 (Bell) and 13 (Rolling stock) are removed. 2 and 7 moved into the Menu dialog. 15 (Trains tabs) is replaced by carousel arrows + selection on cards/map. 18 (per-card mini-maps) is replaced by the Map panel (row M in the table).

3 · Mapping table (element by element)

#	Current prototype	New UI	Notes / action
1	Header: N160 logo + 'SerieN Autumn Railways' + 'N SCALE RAILWAY · 150×70 CM'	Wood plaque 'SERIE N' + brass edition plate (EDITION config)	Logo and subtitle REMOVED
2	Header: Quality select 'Auto · High'	Menu dialog > Graphics quality (Auto/Low/High)	emit setQuality; toggle .q-low
3	Header: '● WEB SIMULATION' status	Annunciator lamp 'Running' (+ Collision/Points/Buffer lamps)	from simulation.paused / emergencyStop
4	Header: Home	Menu button (#menuB) → Menu dialog	old Home = main menu
5	Header: Track plan	Track plan button (#tpB) → Track plan dialog	now interactive
6	Header: ? Help	Help button (#hlB) → Help dialog	
7	Header: Esci	Menu dialog > 'Exit to title' (#exitB)	emit exit
S	(new)	Header: Sound On/Off (#snd)	mirrors Menu > Sound
8	View: 'Bell: on' button	REMOVED	bell only as alarm sound; sound toggle in header
9	View: NEXT SWITCH card (diagram, 'Expected exit', collapse –)	HUD plate #nx (one line; expands on click or < 500 mm)	diagram REMOVED; Straight/Diverging control added
10	View: camera bar (Route, View select, Zoom ±, Pan, Orbit, Home, Follow)	#cam: Zoom+, Zoom–, Recenter always; 'View' expands views + Pan/Orbit/Follow/Route	View select → segmented; old Home → Recenter; auto-collapse 8 s
11	Right: INFRASTRUCTURE / Switches grid with descriptions + paragraph	Switches panel: one tile per switch = ID + Straight|Diverging	generate from game registry; paragraph REMOVED; counter = diverging/total
12	Right: ENVIRONMENT 'Daylight' + slider + Day/Sunset/Night	Environment panel: slider + sun/sunset/moon labels	drives UI lighting + emit setTimeOfDay
13	Right: Rolling stock select 'DB · original models'	REMOVED	not needed with the new train management
14	Right: RAILWAY OPERATIONS / Missions / First lap	Missions panel (compact) with progress bar + Next mission	Next enabled only when complete
M	(new)	Map panel: track map + train markers + predicted route + Run KPI per train	KPI: time & distance since last start
15	Dock: 'Trains' tab bar BR 055/216/260	REMOVED	select via card / marker / key row / keys 1–8; carousel arrows
16	Dock: + Add train, STOP ALL, Pause, Reset	Control cluster: + Train, Pause/Resume, Reset (hold 900 ms), STOP ALL mushroom (latching)	STOP ALL now latches; second press releases
17	Card: loco image	Silhouette <use href=#l1|l2|l3> by train.kind + colour dot	
18	Card: mini-map per train	REMOVED	replaced by the Map panel
19	Card: 'Stopped · 3 wagons · 315 mm'	Status pill + caption 'type · era · wagons · length'	
20	Card: Speed slider + %	Gauge (actual speed) + throttle slider (commanded) + %	inertia; sub-dial BAR/RPM
21	Card: Reverse, Stop, Lights, Uncouple, Couple	Same five; icon buttons with small labels; Lights shows on-state	titles show shortcuts

#	Current prototype	New UI	Notes / action
22	3D layout view	Unchanged (engine). Put the engine canvas inside .win	the UI package does not include the 3D view

Removed features — what to do in the code

- **Bell button**: delete the button and its state. The bell sound is not played anymore except as part of alarms.
- **Rolling stock select**: delete the select and its handler. Train models come from the new train management.
- **Trains tab bar**: delete. Selection = click on card, map marker, map key row, or keys 1–8.
- **Mini-map in each train card**: delete. All positions are shown in the Map panel.
- **Next-switch diagram & explanatory paragraphs**: delete.
- **Logo N160 and subtitle**: delete; replaced by the wood plaque + edition plate.

4 · Layout

The root `<main id="st" class="st">` is a CSS grid over the full viewport (`100dvh`), padding 12 px, gap 10 px:

```
columns: minmax(0,1fr) var(--sw)       --sw =
clamp(250px, 21vw, 330px)
rows:    var(--hh) minmax(0,1fr) var(--dh)
         --hh = clamp(58px, 7.4vh, 72px)   --dh =
156px (146px if height ≤ 800)
areas:   "h h" / "v s" / "d d"
```

Every area is a **module** `.m`: 4 px brushed-brass frame (radius 14) around an enamel panel `.mi` (radius 10) with four round rivets drawn as background images. Modules never move on hover.

Side column (DOM order m-sw, m-env, m-ms, m-lg; visual order via CSS `order`): Environment → Switches → Map → Missions. The Map panel takes the remaining height.

Dock: carousel of train cards (3 visible on desktop), arrows left/right, control cluster on the right, version label bottom-right.

Breakpoint	Behaviour
heightCompact1000	max-height:1000px → mission description hidden; switch descriptions hidden
heightCompact860	max-height:860px → rivets 8px; mission row compact
heightCompact800	max-height:800px (and width>=701) → dock 146px; Missions title hidden; Day/Sunset/Night labels icons only
width1400	max-width:1400px → edition plate letter-spacing reduced
width1300	max-width:1300px → annunciator lamps without text
tablet	max-width:1180px → side 210–260px; header compacted; 2 train cards visible
tabletPortrait	max-width:1100px and portrait → single column: header / view / side panels / dock
phone	max-width:700px or max-height:480px → stacked page, bottom nav (Trains/Switches/Environment/Missions/STOP ALL), 1 train card visible
touch	pointer:coarse → min 44px targets, larger slider knobs

Responsive reference



1280 × 800 — switches compact (no descriptions), mission row compact.



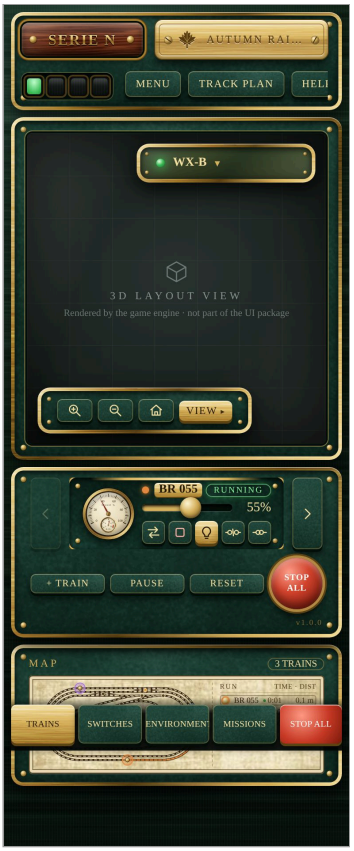
1024 × 768 — 2 cards, side 210–260 px, icon-only phase labels.



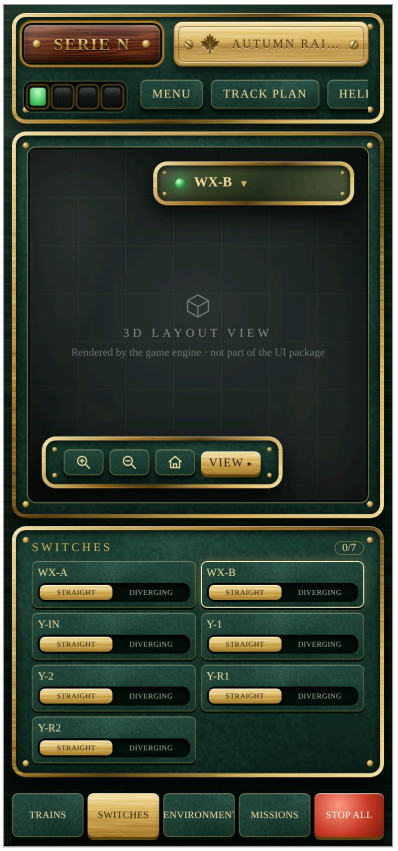
1920 × 1080 — switch descriptions + mission text visible.



Tablet portrait — single column, side panels 2 × 2.



Phone — stacked; bottom nav with STOP ALL.



Phone · Switches tab.

5 · Components

Header



Wood plaque "SERIE N" (static). **Edition plate**: brass, two screws, emblem + name from `EDITION = {name:'Autumn Railways', emblem:'maple'}` — the only thing that changes per edition. **Annunciator**: four lamps (Running · Collision · Points · Buffer), click = acknowledge + select train. **Buttons**: Menu, Track plan, Help, Sound.

Side panels



Environment: range 0–100 (`#day`), labels Day/Sunset/Night with icons; current phase highlighted; drives UI lighting and emits `setTimeOfDay`.

Switches: counter *diverging/total*. One tile per switch: ID (+ description on tall screens) and a Straight | Diverging segmented control, identical to the HUD control. Tile states: `.dev` diverging, `.nx1` next switch of selected train (soft pulse), `.ck` click wobble.

Map: track (sepia + dashed), switch dots (amber when diverging), one brass marker per train with its colour halo, predicted route of the selected train. Key on the right: per train *run time* and *distance* since the last start (reset on 0 → moving, frozen when stopped/paused/E-stop). Scrolls above 4 trains.

Missions: title, progress bar, label, *Next mission* disabled until complete; complete = green bar + chime.

5 · Components (continued)

Train dock



Card (`article.tr`): gauge (left), colour dot + loco silhouette (`#l1` steam, `#l2` diesel, `#l3` shunter) + name plate + status pill, caption, throttle slider + %, five icon buttons (Reverse, Stop, Lights, Uncouple, Couple). **Selected card** (`.sel`): double brass ring, sheen bar on top, name in a brass plate; other cards dimmed to 80 %. **Gauge**: main dial = *actual* speed (needle with overshoot); sub-dial = boiler pressure BAR (steam, cream face) or RPM (diesel: black face; shunter: cream). **Slider** = *commanded* speed; actual speed follows with inertia (24 %/s up, 60 %/s down, 240 %/s under emergency stop).

Control cluster: + Train (max 8) · Pause/Resume · **Reset = hold 900 ms** (red fill; short press does nothing) · **STOP ALL mushroom** latches (stays pressed, label RELEASE, throttles locked); press again to release (twist animation).

View overlays



HUD expanded (top right) and camera bar expanded (bottom left).

Next-switch HUD `#nx`: one line (lamp, switch ID, distance). Expands on click or automatically when the selected moving train is < 500 mm away (amber lamp + warning tone). Expanded: train, description · position, Straight|Diverging (same command as the tile). Hidden with no trains.

Camera bar `#cam`: Zoom +, Zoom −, Recenter always visible; *View* ▸ reveals views (Overview, Station, Depot, Loop), Pan/Orbit (exclusive), Follow, Route (toggles). Auto-collapses after 8 s.

Alarm slip `#slip`: slides down at top centre with lamp, title, detail and actions; shows the most severe alarm, "+N" for more.

6 · Component reference (selectors & states)

Component	Selector	Details
woodPlaque	`.hd .plq > b.inl`	**content** Static text 'SERIE N' **notes** Mahogany texture, two brass rivets .rvp. Height equals edition plate.
editionPlate	`#edp`	**content** #edn = EDITION.name; <use> emblem = '#emb-'+EDITION.emblem **config** EDITION = {name:'Autumn Railways', emblem:'maple'} **notes** Two slotted screws .scr. Swap name/emblem per edition; nothing else changes.
annunciator	`.ann > button.anl[data-c]`	**channels** **sim**: Running/Paused/E-stop; **col**: Collision; **pts**: Points; **buf**: Buffer **states** **ok**: green (sim only); **wn**: amber steady; **al**: red; **blk**: added to 'al' when unacknowledged: blinks 1.1 s; **(none)**: off **click** acknowledge all alarms of that channel + select involved train
headerButtons	`.hc`	**items** **#menuB**: open Menu dialog; **#tpB**: open Track plan dialog; **#hlB**: open Help dialog; **#snd**: toggle sound; label 'Sound: On|Off'; mirrors Menu > Sound
nextSwitchHud	`#nx`	**collapsed** #nxh row: lamp, #nxn switch id, #nxd distance mm **expanded** #nxl train name, #nxe 'description · straight|diverging', #nxs Straight|Diverging segmented **expandWhen** user click (toggle .open) OR (distance<500 mm AND selected train moving) → class .near (amber lamp, warn sound once) **hiddenWhen** no trains (.gone) **binding** first facing-point switch ahead of the SELECTED train on its current route **action** #nxs buttons set that switch (same as tile)
alarmSlip	`#slip`	**binding** highest-priority active alarm (al > wn); 'in' items never shown **parts** **#slt**: title; **#sld**: detail (mm); **#sln**: '+N' more; **#sla**: action buttons **show** add .on; add .al for red **actions** from alarm.actions + 'Show' (select train) + 'Ack' for red alarms
cameraBar	`#cam`	**alwaysVisible** Zoom in, Zoom out, Recenter **toggle** #camt 'View' toggles .open; #cmo is inert when closed **expanded** **views**: ['Overview', 'Station', 'Depot', 'Loop']; **Pan / Orbit**: mutually exclusive (data-c); **Follow / Route**: independent toggles (data-k) **autoCollapseMs** 8000 **emits** camera
environmentPanel	`.m-env`	**controls** **#day**: range 0..100; set --p custom property = value%; **#dl**: 'Day'|'Sunset'|'Night' (v<33, v<66, else); **.lab span.on**: highlight current phase **emits** setTimeOfDay **lighting** see 'lighting'
switchesPanel	`.m-sw`	**counter** #cnt = diverging/total **tile** **markup**: IDdescription <button data-v="0" class="on" aria-pressed="true">Straight</button><button data-v="1" aria-pressed="false">Diverging</button> ; **states**: {'.dev': 'switch diverging', '.ck': 'click wobble 280 ms', '.nx1': 'is the next switch of the selected train (soft pulse)', 'button.on': 'current position (aria-pressed)'}; **grid**: 2 columns; N tiles generated from the game switch registry (NOT hard-coded) **emits** setSwitch
mapPanel	`.m-lg`	**counter** #mpc 'N train(s)' **map** svg.map: <use href=#lay> track twice (sepia + dashed); #md switch dots (.mdot, .dv amber); #mm train markers (.tm; .s selected pulses; .wn/.al); #mr predicted route of selected train in its colour **key** #mkl rows button.mkr: brass dot (halo = train colour), name, .k1 run time, .k2 distance; .on selected; .run moving (green dot); .al alarm **emptyState** .mkE 'No trains' **click** marker or row selects train
missionsPanel	`.m-ms`	**fields** **#msn**: title; **#msd**: description (hidden below 1000 px viewport height); **#mpf**: progress fill width %; **#mpt**: 'x.x / y.y m' or status; **#mnx**: Next mission; disabled until progress = 1; panel gets .dn when complete (green bar, chime once); **h2 em**: index/total

Component	Selector	Details
trainCarousel	`#trw > #trk`	**arrows** #aL/#aR scroll by one card; disabled at ends; hidden when no trains **cardWidth** 3 per row desktop, 2 below 760 px container, 1 below 430 px; cards grow when fewer trains (max half width) **emptyState** .trE with '+ Add train'
trainCard	`article.tr`	**states** **.sel**: selected (double brass ring, top sheen bar, brass name plate); **not .sel**: dimmed (brightness .8); **.alm**: red pulsing ring; **.wrn**: amber ring; **.lk**: throttle locked (E-stop) **parts** **gauge**: svg.gg from gauge(train): main SPEED % needle .nd shows ACTUAL speed; sub-dial .nd2 = BAR (steam) or RPM (diesel/shunter); **nameRow**: .tg colour dot, svg.lc <use href=#l{kind}>, h3 name, .pl status pill; **pill**: {'Stopped': 'neutral', 'Running': '.go', 'Reversing': '.bk', 'alarm text': '.al / .wn'}; **caption**: .cap '{type} · {era} · {wagons} · {length}'; **throttle**: input.thr 0..100 = COMMANDED speed; .pc percent; **buttons**: [data-a] rev, stop, lt (.on when lights on), unc, cpl — icon + small label **emits** setSpeed, reverse, stopTrain, lights, couple, uncouple, selectTrain
controlCluster	`.cc`	**items** **[data-g=add]**: + Train (max 8); **[data-g=pause]**: Pause ↔ Resume (label swaps); **#rst**: Reset: HOLD 900 ms (red fill, custom property --h 0..1); early release cancels; keyboard: hold Space/Enter while focused; **.mush**: STOP ALL mushroom: press = latch E-stop (.lt2, label 'RE LEASE'); press again = release (.tw twist animation)
versionLabel	`#ver`	**content** 'v'+VERSION, bottom-right of dock
mobileNav	`nav.nav`	**visibleAt** phone breakpoint **tabs** **treni**: Trains (dock + map); **scambi**: Switches; **ambiente**: Environment; **missioni**: Missions **binding** #st[data-tab] **stop** STOP ALL mirror
menuDialog	`#dlg`	**sections** **#qsg**: Graphics quality Auto/Low/High + #qd description + #fps live; **#ssg**: Sound On/Off **actions** **#exitB**: Exit to title (emits exit); **#dlgC**: Resume (close) **close** Esc, click on backdrop, Resume; focus returns to the opener
trackPlanDialog	`#tpD`	**content** Large parchment track plan; switch labels .tpl (amber when diverging); click toggles switch; #tpC counter
helpDialog	`#hlD`	**content** Keyboard list + annunciator legend (static)

7 · States



Collision alarm: annunciator red blinking, slip with actions, card ring red, pill "Too close".



Emergency stop latched: mushroom pressed (RELEASE), Running lamp red, throttles locked.



Sunset: warm wash (canvas, multiply), not over the 3D view.



Night: panels dark; gauges and controls lit by hidden bulbs (CSS only, `--n`).



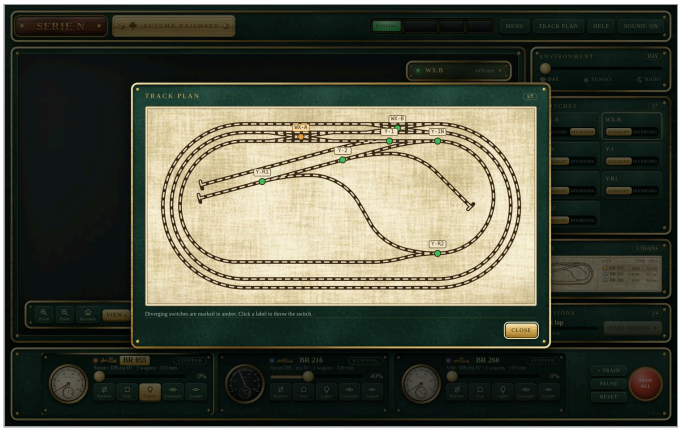
0 trains: dock empty state, key "No trains", HUD hidden.



8 trains: carousel arrows, key scrolls.



Low quality: flat materials, no blur, no decorative motion.



Track plan dialog: click a label to throw a switch.

7 · States (rules)

Alarm rules

Channel	Level	When	Title	Actions
col	wn	moving train, another train ahead within 450 mm	Collision risk	Stop <train>
col	al	distance <= 100 mm; train auto-held at 90 mm (emergency brake)	Emergency brake	Reverse, Show, Ack
pts	wn	moving train will enter a switch from the trailing side set against it, within 450 mm	Points set against	Set <id>
pts	al	train held at that switch (<= 3 mm)	Points set against	Set <id>, Show, Ack
buf	wn	moving train within 350 mm of a buffer stop	Buffer stop ahead	Stop <train>
buf	in	train stopped at buffer (<= 3 mm)	At buffer stop	
sim	al	emergency stop latched	Emergency stop	Release

Levels: al red, blinking until acknowledged; wn amber steady; in info (lamp only, no slip). New *al* plays the alarm bell, new *wn* the warning tone. Ack (button or annunciator click) only stops the blinking; the alarm disappears when its condition clears.

Lighting

Input `timeOfDay` v (0–100). Sunset factor `LK = max(0, 1 − |v − 50| / 34)`, night factor `LN = clamp((v − 42) / 50, 0, 1)`. Set `--n = LN` and `--k = LK` on `#st`; add class `nt` when LN > 0.5. Night is entirely CSS. Sunset uses `canvas.lx` created by `light()` (disabled in Low). The 3D scene lighting is the engine's job (send `setTimeOfDay`).

Graphics quality (Menu)

Auto / Low / High, persisted in `localStorage['sn-q']`. Auto = Low if ≤ 4 CPU cores, ≤ 4 GB memory or reduced-motion; switches to Low if fps < 40 after 5 s. Low adds `.q-low` on `#st`: textures off, no blur, no decorative animation, 30 fps UI loop. Alarms, sounds and all functions stay.

Keyboard

Key	Action
`Space`	STOP ALL / release (ignored when a button has focus)
`1-8`	select train
`ArrowUp/ArrowDown`	selected train speed +/-5 (blocked during E-stop)
`R`	reverse
`S`	stop selected
`L`	lights
`M`	menu
`Esc`	close dialog
`ignore`	when focus is in a dialog, when a range input uses arrows, with Ctrl/Meta/Alt

Sound

All sounds are synthesised with Web Audio (object `SFX` in the reference — port it as-is, no audio files). The AudioContext starts on the first user gesture. Persisted in `localStorage['sn-snd']`. Rail-joint clicks every 124 mm of real distance, panned by position, quieter for non-selected trains.

8 · Integration with the game

Where things go

- **3D canvas**: inside `.win` as first child, `position:absolute; inset:0`. Delete `.vph` and `svg.lay3d`. Keep `#nx`, `#slip`, `#cam` after it. Only overlays capture pointer events.
- **Sprite**: inline `assets/svg/serien-sprite.svg` once at the top of `<body>` (fix the image path of pattern `#spun` to where the textures are served). Fill `<g id="lay">` with the track polylines at runtime.
- **Track geometry** for Map and Track plan: edges as polylines in layout units (reference: 4022 × 1916 units = 150 × 70 cm).

State model (game → UI)

```
{
  "$note": "The UI is a VIEW of this state. Produce it from the game every tick (or on change) and call the
render functions. Field names are a recommendation; keep them stable.",
  "simulation": {
    "paused": "boolean",
    "emergencyStop": "boolean"
  },
  "environment": {
    "timeOfDay": "number 0..100"
  },
  "selectedTrainId": "string | null",
  "trains": [
    {
      "id": "string (stable)",
      "name": "'BR 055'",
      "kind": "1 steam | 2 diesel | 3 shunter (selects silhouette l{kind} and gauge type)",
      "type": "'Steam'",
      "era": "'DB era III'",
      "wagons": "int",
      "lengthMm": "int",
      "colorIndex": "0..7 (index into tokens color.train.list)",
      "speedCmd": "0..100 (slider)",
      "speedActual": "0..100 (needle, motion)",
      "direction": "1 | -1",
      "lights": "boolean",
      "subDial": "pressure bar (steam) or rpm (diesel) — optional, UI can derive from speedActual like the
reference",
      "run": {
        "timeS": "number",
        "distanceMm": "number"
      },
      "position": {
        "x": "layout units",
        "y": "layout units"
      },
      "nextSwitch": "{id, distanceMm} | null",
      "predictedRoute": "[[x,y],...] in layout units (about 2600 units ahead)",
      "status": "Stopped | Running | Reversing | <alarm text>"
    }
  ],
  "switches": [
    {
      "id": "string",
      "description": "string",
      "diverging": "boolean"
    }
  ],
  "alarms": [
    {
      "key": "stable string",
      "channel": "col|pts|buf|sim",
      "level": "al|wn|in",
      "title": "string",
```

```
      "detail": "string",
      "trainId": "string | null",
      "actions": [
        {
          "label": "string",
          "event": "ui event object"
        }
      ]
    }
  ],
  "missions": {
    "index": "int",
    "total": "int",
    "title": "string",
    "description": "string",
    "progress": "0..1",
    "progressLabel": "string",
    "complete": "boolean"
  },
  "layoutGraph": {
    "$note": "Track geometry in layout units for Map and Track plan. Reference uses PDF units (4022 x 1916 for
150 x 70 cm). Provide from the game's track model.",
    "edges": "[{pts:[[x,y],...]}]",
    "bufferStops": "[[x,y],...]",
    "switchPositions": "{id: [[x,y],...]}"
  }
}
```

8 · Integration (continued)

Events (UI → game)

Dispatch `window.dispatchEvent(new CustomEvent('serien:ui', {detail:{type, ...payload}}))` (already used for `exit`). Any bus is fine; keep the payloads.

type	payload
`setSwitch`	{"id": "string", "diverging": "boolean"}
`setSpeed`	{"trainId": "string", "value": "0..100"}
`reverse`	{"trainId": "string"}
`stopTrain`	{"trainId": "string"}
`lights`	{"trainId": "string", "on": "boolean"}
`couple`	{"trainId": "string"}
`uncouple`	{"trainId": "string"}
`selectTrain`	{"trainId": "string"}
`addTrain`	{}
`pause`	{"on": "boolean"}
`reset`	{}
`emergencyStop`	{"on": "boolean"}
`setTimeOfDay`	{"value": "0..100"}
`camera`	{"action": "zoomIn|zoomOut|recenter|pan|orbit|follow|route|view", "value": "view name or boolean"}
`ackAlarm`	{"keys": "string[]"}
`nextMission`	{}
`setQuality`	{"value": "auto|low|high", "effective": "low|high"}
`setSound`	{"on": "boolean"}
`exit`	{}

Reference JavaScript: what to keep, adapt, replace

KEEP port as-is: $, SFX, railJ (feed real distance), fmtT, gauge, GS, pt, applyQ, QD, paintS, openM / closeM, openD / closeD, setCam, arr, go, emptyUpd, hold-to-reset block (startH / stopH / doReset → emit reset), keyboard handler, light (sunset canvas), estopOn / release (emit emergencyStop), slip, annunciator click handler.

ADAPT keep the rendering, feed it from game state: mk (create card + markers + key row from a train), upd / all (render trains from state), sets (render switch + emit setSwitch), pick (emit selectTrain), routeUpd (draw #mr from train.predictedRoute; HUD from train.nextSwitch), scan (render state.alarms instead of computing), msUpd (render missions from state), place (markers from train.position), tpUpd (labels from switchPositions).

MOCK delete and replace with game data/logic: G (track graph JSON), walk, at, nearest, nextE, predict, scanAhead (move into the game if needed), spawn / SPAWN, NEW (sample trains), T initial array, MS missions, requestAnimationFrame movement loop (speedActual integration, KPI accumulation, totD), LAY / MT / toP / toPh / scenery (old 3D placeholder, already hidden).

Speed: two values, always

`speedCmd` (slider, what the player asked) and `speedActual` (needle, movement, KPI). The pill "Running" and the KPI reset use `speedCmd` going from 0 to > 0. Emergency hold (collision/points) sets both to 0 immediately.

9 · Step-by-step plan for Sol

Step 0 — Explore (no code)
Open the reference and the library. Reproduce every state in section 7. Read `spec/ui-spec.json` once.

Step 1 — Assets
Copy `tokens/`, `css/`, `assets/` into the game's static folder keeping the relative paths (`css/` must reach `../assets/textures/`). Self-host the three fonts or keep the Google Fonts @import. *Check:* no 404 in the network panel.

Step 2 — Shell markup
Replace the old UI DOM with the reference DOM (copy from `<main id="st">` to the end of the dialogs). Remove the old header, side panel, dock, Bell, Next-switch card, camera bar, tabs and selects. Put the engine canvas inside `.win`. *Check:* at 1440 × 900 the page matches `01-desktop-day.jpg` with the mock data.

Step 3 — Port KEEP modules
Move the reference `<script>` into a module. Keep the KEEP functions. *Check:* dialogs, sound, quality, keyboard, hold-reset, STOP ALL latch still work with the mock.

Step 4 — State adapter
Write `toUiState(gameState)` returning the model of section 8. Replace the mock arrays (T, SW, MS) and the movement loop with this state, re-rendering through `mk / upd / sets / routeUpd / msUpd / scan` (ADAPT). *Check:* with the real game: trains, switches, map and KPI update; `?trains` no longer needed.

Step 5 — Events
Emit the events of section 8 from every control and handle them in the game. Switch commands must update tile, HUD, Track plan and map dot from the returned state (not optimistically from the click). *Check:* every row of the mapping table.

Step 6 — Alarms
Either compute alarms in the game (preferred) using the rules of section 7, or port `scanAhead/scan`. Render with `slip` + annunciator. *Check:* each rule triggers lamp, slip, card, marker, sound; Ack stops blinking.

Step 7 — Responsive & quality
Test 1920 × 1080, 1440 × 900, 1280 × 800, 1024 × 768, 820 × 1180, 390 × 844; Low/High; reduced motion.

Step 8 — Clean-up
Delete all MOCK code, the placeholder, and unused old CSS. Set `VERSION` from the build. Run the acceptance checklist.

Common mistakes to avoid

- Reordering/merging CSS rules → breaks responsive and night mode.
- Putting `transform`/hover lift on `.m` modules → they are riveted, they never move.
- Sheen on non-interactive elements → only buttons and switch controls have hover sheen.
- Using the slider value for the needle → needle shows actual speed.
- Hard-coding 3 trains / 7 switches, or the mock switch IDs.
- Toggling switches optimistically on click → render from game state.
- Making Reset a simple click → must be a 900 ms hold.
- Making STOP ALL momentary → it latches until released.
- Forgetting `@property --n` → night transition snaps instead of fading.
- Letting `.win` capture clicks → the 3D view must receive them.

10 · Acceptance checklist

1. Visual check at 1440×900 against the PDF screenshots (with fonts loaded): layout, spacing, colours match.
2. 0, 1, 2, 3, 5, 8 trains: no overflow, carousel arrows correct, empty states at 0.
3. Selected train highlighted on card, map marker and key row; keys 1–8 select.
4. Switch thrown from tile, HUD and Track plan: all three + map dot update; counter correct.
5. STOP ALL latches (throttles disabled, annunciator red); second press releases.
6. Reset requires a 900 ms hold; a short press does nothing.
7. Each alarm rule produces lamp, slip, card ring, marker colour and sound; Ack stops blinking.
8. Night: panels dim, gauges/controls lit by hidden bulbs, ~1.2 s transition; no glow pools.
9. Low quality: no textures/blur/decorative animation; alarms still blink.
10. 1024×768, 1280×800, 1920×1080, 390×844: nothing clipped; side column fits; phone bottom nav works.
11. Keyboard: every shortcut works; Esc closes dialogs; focus ring visible (2 px #fff3c4).
12. prefers-reduced-motion: no animation.

Behaviour constants (tokens.json → behaviour)

```
{
  "trains": {
    "min": 0,
    "max": 8,
    "visibleCardsDesktop": 3,
    "visibleCardsTablet": 2,
    "visibleCardsPhone": 1
  },
  "speed": {
    "stepKeyboard": 5,
    "accelPctPerSec": 24,
    "brakePctPerSec": 60,
    "emergencyBrakePctPerSec": 240,
    "usage": "slider = commanded speed; needle/motion = actual speed"
  },
  "alarms": {
    "collisionWarnMm": 450,
    "collisionHoldMm": 90,
    "collisionAlarmMm": 100,
    "pointsWarnMm": 450,
    "pointsHoldMm": 3,
    "bufferWarnMm": 350,
    "bufferAtMm": 3,
    "scanHz": 6
  },
  "hud": {
    "autoExpandWithinMm": 500,
    "autoExpandOnlyIfMoving": true
  },
  "camera": {
    "autoCollapseMs": 8000
  },
  "resetHoldMs": 900,
  "switchMotionMs": 520,
  "kpi": {
    "timeFormat": "m:ss, h:mm:ss after 1h",
    "distanceUnit": "m, 1 decimal",
    "resetsWhen": "speed goes 0 → >0",
    "freezesWhen": "stopped, paused, emergency stop"
  },
  "quality": {
    "auto": "low if hardwareConcurrency<=4 or deviceMemory<=4 or prefers-reduced-motion; switches to low if
fps<40 after 5s",
    "lowFps": 30,
    "persist": "localStorage sn-q"
```

```
  },
  "sound": {
    "masterGain": 0.6,
    "railJointMm": 124,
    "persist": "localStorage sn-snd"
  }
}
```

Colour tokens (excerpt)

`100` #f6e2a4 `300` #d9b45f `500` #a9802f `700` #5f4416 `screw` #4a3410 `specular` #fff7d6
`go` #3fbf5a `goGlow` #4cff7c `warn` #eca23a `alarm` #dc432e `stopButton` #d4452f
`primary` #f2e8d0 `secondary` #a9bfae `onBrass` #2a1d08 `nightLabel` #c99a52

Full list: `tokens/tokens.json`, CSS variables `--sn-*` in `tokens/tokens.css`.